

May 6, 2026

11:49:52 AM

Starting parallel pool (parpool) using the 'Processes' profile ...

Connected to parallel pool with 12 workers.

Copying objective function to workers...

Done copying objective function to workers.

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Iter	Active	Eval	Objective	Objective	BestSoFar	BestSoFar	SectionDepth	InitialLearn-
Momentum	L2Regulariza-	tion	workers	result	runtime	(observed)	(estim.)	Rate
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1	1	12	Best	0.762	101.33	0.762	0.762	3	0.71501
0.86017		0.0084489							
2	12	Accept	0.762	139.7	0.762	0.762	0.762	2	0.25803
0.94448		2.9357e-09							
3	12	Accept	0.788	151.53	0.762	0.76328	0.762	3	0.24648
0.96098		7.554e-07							
4	12	Accept	0.8	47.925	0.762	0.762	0.762	2	0.55566
0.91646		3.1756e-05							
5	12	Accept	0.848	194.2	0.762	0.76474	0.762	3	0.47279
0.87449		2.6556e-05							
6	12	Accept	0.848	44.584	0.762	0.77258	0.762	2	0.54545
0.94437		6.1478e-09							
7	12	Accept	0.848	21.249	0.762	0.76387	0.762	1	0.82014
0.91072		1.2495e-09							
8	12	Accept	1	31.574	0.762	0.76201	0.762	1	0.77427
0.87413		0.0079127							
9	12	Best	0.084	331.73	0.084	0.10908	0.084	1	0.071244
0.83028		3.408e-06							
10	12	Accept	0.102	353.52	0.084	0.082598	0.084	1	0.10281
0.87562		1.741e-05							
11	12	Accept	0.848	376.49	0.084	0.088329	0.084	2	0.21196
0.94114		1.3204e-06							

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25	12	Accept	0.124	296.22	0.084	0.0909	1	0.076197	✓
0.97595	7.9368e-07								
26	12	Accept	0.104	326.04	0.084	0.093875	1	0.079571	✓
0.97672	1.0966e-10								
27	12	Accept	0.204	307	0.084	0.093792	1	0.01007	✓
0.8029	0.0067004								
28	12	Accept	0.186	291.84	0.084	0.09242	1	0.025266	✓
0.80312	0.0042415								
29	12	Accept	0.104	279.37	0.084	0.092153	1	0.03307	✓
0.80728	2.2993e-10								
30	12	Accept	0.098	463.44	0.084	0.095241	2	0.011894	✓
0.80362	1.0896e-10								

Optimization completed.
MaxObjectiveEvaluations of 30 reached.
Total function evaluations: 30
Total elapsed time: 1164.3345 seconds
Total objective function evaluation time: 8917.029

Best observed feasible point:

SectionDepth	InitialLearnRate	Momentum	L2Regularization
1	0.071244	0.83028	3.408e-06

Observed objective function value = 0.084
Estimated objective function value = 0.14503
Function evaluation time = 331.7348

Best estimated feasible point (according to models):

SectionDepth	InitialLearnRate	Momentum	L2Regularization
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1          0.076939      0.94998      5.9551e-10
```

```
Estimated objective function value = 0.095241
Estimated function evaluation time = 332.1646
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```
acc_25dB =
```

```
0.9314
```

```
acc_20dB =
```

```
0.7672
```

```
acc_15dB =
```

```
0.5292
```

```
acc_10dB =
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```
0.2836
```

```
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```